

**Hedera Hashgraph as a Scalable Distributed Ledger: Addressing Blockchain Scalability
and Consensus Limitations**

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Abstract

Blockchain and distributed ledger technologies (DLTs) have revolutionized decentralized systems but face significant hurdles in scalability, energy efficiency, and transaction throughput, primarily due to sequential block-based consensus mechanisms like Proof-of-Work (PoW). This paper investigates Hedera Hashgraph as a next-generation DLT that addresses these limitations by utilizing a Directed Acyclic Graph (DAG) architecture and an Asynchronous Byzantine Fault Tolerant (aBFT) consensus protocol. By employing a gossip protocol and virtual voting, Hedera enables high-throughput, low-latency transaction processing without the need for sharding or off-chain layer-2 solutions. This study analyzes the architectural transition from traditional blockchains to DAG-based ledgers, highlighting how Hedera achieves deterministic, leaderless consensus in real-world asynchronous environments. The findings indicate that Hedera offers a robust, secure, and energy-efficient alternative for modern distributed applications, providing strong finality guarantees where traditional probabilistic models fail.

Keywords: Distributed Systems, Scalability, aBFT, Hedera Hashgraph, DAG, Consensus, Fault Tolerance

1. Introduction

Distributed ledger technologies (DLTs) have fundamentally altered the paradigm of trust, decentralization, and data integrity in contemporary computational infrastructure (Bano et al., 2019). By establishing shared, cryptographically verified data stores replicated across mutually distrusting nodes, distributed ledgers eliminate the single points of failure and systemic vulnerabilities inherent in centralized frameworks (Zheng et al., 2017).

The structural evolution of distributed systems has progressed from early linear blockchains to advanced Directed Acyclic Graph (DAG) frameworks, reflecting a persistent effort to optimize performance without sacrificing architectural security (Wang et al., 2020). Despite their transformative potential, traditional linear blockchains remain constrained by fundamental scalability bottlenecks. The necessity of global block ordering, linear block sequencing, and high-overhead consensus mechanisms natively restricts transaction throughput while inflating operational latency and processing costs (Vukolić, 2015).

To systematically evaluate how modern distributed architectures address these limitations, this paper analyzes the design and mechanics of Hedera Hashgraph. Hedera represents a distinct paradigm shift away from traditional block-based serialization. By replacing the sequential chain of blocks with a graph-structured Directed Acyclic Graph (known as the hashgraph) and utilizing an Asynchronous Byzantine Fault Tolerant (aBFT) consensus protocol, Hedera demonstrates how parallel transaction processing and message-efficient voting mechanisms can mitigate performance degradation at scale (Baird, 2016).

This study examines the core structural properties, network protocols, and mathematical guarantees that distinguish Hedera Hashgraph from legacy decentralized platforms, presenting an architectural assessment within the broader framework of modern distributed systems design.

2. Literature Review

The development of distributed ledger architectures is defined by an ongoing pursuit to resolve the scalability trilemma, which states that a decentralized system can concurrently satisfy only two of three core properties: security, decentralization, and scalability (Buterin, 2017). In legacy platforms, the primary performance inhibitor stems from the structural coupling of data storage and consensus sequencing.

Traditional Proof-of-Work (PoW) architectures, such as Bitcoin, achieve consensus through computational competition (Nakamoto, 2008). Nodes compete to solve cryptographically intensive puzzles to secure the right to append a highly localized bundle of transactions (a block) to a singular, universally shared chain. This architecture establishes an adversarial bottleneck: to prevent catastrophic chain splits (forks), the network's block generation rate must be artificially restricted, enforcing low throughput and extended probabilistic finality windows (Vukolić, 2015). Under probabilistic finality, a transaction is never absolute; its security instead increases exponentially as subsequent blocks are appended, making it highly sensitive to network latency.

To alleviate the extreme energy expenditure and throughput constraints of PoW, alternative frameworks introduced Proof-of-Stake (PoS) and leader-based architectures (King & Nadal, 2012). While PoS drops computational competition in favor of economic stake weighting, many implementations still rely on a designated leader node or a shifting committee to propose the next state update.

Leader-based distributed systems introduce specific vulnerabilities, such as vulnerability to targeted Distributed Denial of Service (DDoS) attacks aimed at the active leader node, as well as the structural overhead of view-change protocols when a leader exhibits arbitrary or malicious failure (Castro & Liskov, 1999). Furthermore, optimization vectors like network sharding—the horizontal partitioning of states across semi-autonomous subsets of nodes, and Layer-2 off-chain networks introduce significant architectural complexity, cross-shard synchronization latency, and secondary security vectors (Wang et al., 2020).

By contrast, modern distributed systems research focus has shifted toward leaderless,

graph-based configurations. Directed Acyclic Graphs decouple transaction submission from global block creation, enabling nodes to construct and propagate state updates asynchronously and concurrently without awaiting a single coordinator (Baird, 2016).

Hedera Hashgraph addresses these specific gaps by deploying an Asynchronous Byzantine Fault Tolerant (aBFT) protocol that delivers deterministic finality without traditional blocks, heavy computational validation, or localized leader vectors, setting a new benchmark for fault tolerance in open, asynchronous networks (Baird et al., 2019).

3. Methodology

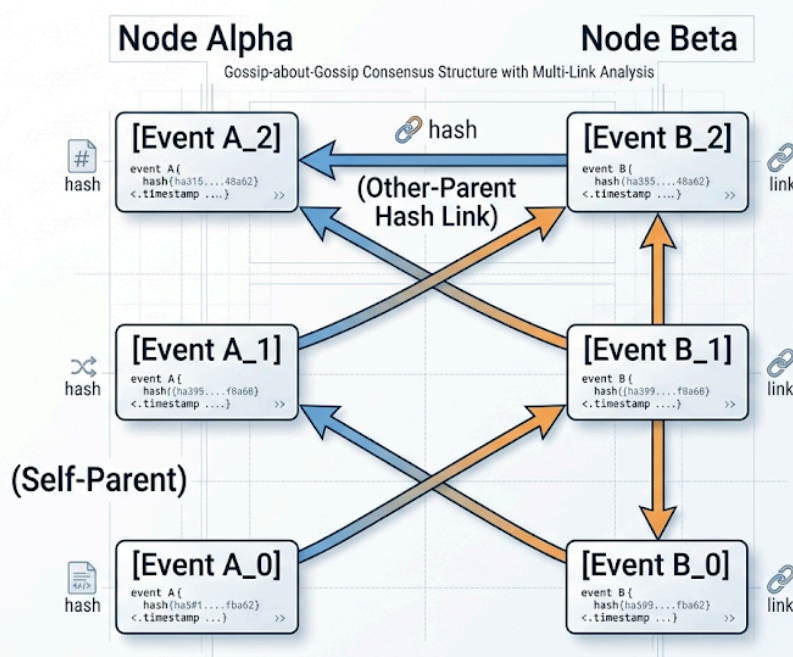
This research evaluates the structural mechanics and performance characteristics of Hedera Hashgraph by breaking down its architectural layers and analyzing its message propagation patterns. The analytical framework targets the elimination of the linear block bottleneck by evaluating data flow across two core layers: the structural graph generation layer (the Hashgraph DAG) and the mathematical agreement layer (the aBFT consensus algorithm) (Baird, 2016).

3.1 Network Topology and Graph Generation

The network topology consists of a set of independent validator nodes operating within an asynchronous environment where network message delays are unbounded but finite. Instead of a single sequence of blocks, the ledger state is captured in a cryptographically validated DAG (Baird, 2016).

Every event node within the hashgraph is defined as a data structure containing:

- A payload of zero or more transactions.
- A timestamp indicating local creation time.
- Two parent cryptographic hashes: the self-parent hash (referencing the producing node's previous event) and the other-parent hash (referencing the latest event received from another node via gossip).



This dual-parent architecture ensures that the graph structure itself mirrors the exact history of network communication, creating an immutable, distributed audit trail of how information moved through the system (Baird et al., 2019).

3.2 Protocol Foundations and Algorithmic Assumptions

The core protocol operates without relying on synchronization assumptions or global network clocks. The mathematical evaluation assumes a standard Byzantine setting where an adversary can control a subset of nodes and manipulate message delivery timing arbitrarily, short of permanently severing network communication (Lamport et al., 1982). The definitive structural constraint governing the network state is defined by the classical Byzantine bound:

$$n > 3f$$

where n represents the total network voting weight (or total node count in an unweighted deployment) and f constitutes the maximum allowable threshold of arbitrary, malicious, or failing entities within the system (Lamport et al., 1982). Under this methodology, the system achieves non-blocking, livelock-free performance across fully asynchronous environments (Baird, 2016).

4. Implementation

Hedera Hashgraph replaces traditional linear sequence verification with two closely integrated mechanisms: the Gossip-about-Gossip protocol for information delivery and Virtual Voting for global transaction ordering (Baird, 2016).

4.1 Gossip-about-Gossip

Information propagates across the network via an optimized gossip protocol. When Node Alpha initiates communication, it randomly selects Node Beta from the active membership pool and transmits its entire localized view of the hashgraph DAG that Node Beta has not yet recorded. Upon receiving this payload, Node Beta instantiates a new event within its local memory (Baird et al., 2019).

This newly initialized event acts as a cryptographic container. It embeds a sequence of new client transactions alongside two structural references: the self-parent hash, pointing to Node Beta's immediately preceding local event, and the other-parent hash, pointing directly to the final event generated by Node Alpha prior to the transmission sync.

Because every event contains the hashes of its parents, nodes do not merely distribute raw transaction data; they distribute a verifiable history of the network's communication topology. This is why the protocol is formalized as Gossip-about-Gossip: nodes are gossiping about the history of the gossip itself (Baird, 2016). The network overhead is highly optimized because nodes only transmit small cryptographic hashes representing graph edges, allowing the underlying Directed Acyclic Graph to expand rapidly and concurrently across all participating nodes without serialization constraints.

4.2 Virtual Voting

The defining characteristic of Hedera's consensus model is Virtual Voting. In standard Byzantine Fault Tolerant systems, nodes must achieve consensus by exchanging a sequence of explicit vote messages across multiple sequential rounds, which creates an $O(n^2)$ or $O(n^3)$ bandwidth

bottleneck that degrades performance as the network expands (Castro & Liskov, 1999).

Hedera eliminates this communication overhead by calculating votes locally. Because every validator node maintains an identical, cryptographically validated copy of the hashgraph DAG, each node possesses complete visibility into exactly when and how every other node received a given transaction (Baird, 2016).

Using the local DAG as an input matrix, a node can run a deterministic voting algorithm to calculate exactly how any other node would vote in a traditional message-passing scheme. Nodes vote virtually by running identical calculations over their local graphs, yielding identical consensus outputs without transmitting a single vote across the network wires (Baird et al., 2019).

The underlying execution flow for virtual voting is governed by structural definitions within the DAG:

- **Round Created:** An event's round R is determined relative to its parents. A starting event is designated as Round 1. For any subsequent event e , its round is initially set to the maximum round of its parents, $R = \max(R_{self_parent}, R_{other_parent})$. However, if e can strongly see a supermajority of "witness" events from round R , its round increment is triggered, setting its round to $R + 1$ (Baird, 2016).
- **Witnesses:** The very first event created by a validator node in any discrete round is designated as a witness. These events serve as the focal points for the voting and round advancement logic.
- **Seeing and Strongly Seeing:** An event $e1$ is said to see an older event $e2$ if there is a valid, downward-directed cryptographic path from $e1$ to $e2$ within the DAG. Crucially, an event $e1$ strongly sees $e2$ if it can trace valid downward paths to $e2$ through a set of intermediate events created by a distinct supermajority of nodes, meaning the aggregate voting weight of these intermediate creators exceeds $\frac{2}{3}$ of the total network weight ($> \frac{2}{3}.n$) (Baird, 2016).

Once a witness event's round is confirmed, the network enters the validation phase. Higher-round

witnesses act as an internal jury to vote on whether lower-round witnesses are officially accepted into the permanent consensus history. If a round $R+1$ witness strongly sees a round R witness, it registers an internal vote of true (Baird, 2016). Because every node runs this identical state machine over its local graph, they all independently arrive at the exact same Byzantine agreement outcome. No network coordination, leader arbitration, or explicit voting phases are required.

4.3 Consensus Timestamping and Ordering

Once the network decides which witness events are officially accepted, it calculates the exact global position and ordering of every individual transaction embedded within the DAG. For any given transaction, its global order is determined by calculating its consensus timestamp.

To determine the consensus timestamp for a specific event e , the system identifies all the witness events in the first round that can see e . The local arrival times of e at each of the validator nodes that generated those witnesses are collected into an ordered set. The network then takes the median value of that set as the definitive, immutable consensus timestamp for event e (Baird, 2016).

$$\text{Consensus Timestamp} = \text{median}\{t_{arrival}^1, t_{arrival}^2, \dots, t_{arrival}^m\}$$

Sorting all transactions chronologically by their consensus timestamps establishes a fair, globally uniform transaction sequence. If two transactions happen to share an identical median timestamp, the tie is broken deterministically by comparing the digital signatures of the events that contain them.

This process delivers Fair Ordering: a malicious node cannot manipulate the final order of transactions by prioritizing its own blocks or front-running user submissions, as the global sequence is determined by the collective median arrival time across the entire network topology (Baird et al., 2019).

5. Results and Discussion

The architectural decoupling of transaction dissemination from global block production yields distinct performance metrics that differentiate Hedera Hashgraph from legacy, linear distributed ledgers.

5.1 Throughput and Latency Profiling

Traditional linear blockchains are bound by serial block confirmation constraints, forcing a performance trade-off between throughput and latency. Bitcoin delivers approximately 7 Transactions Per Second (TPS) with confirmation latencies spanning minutes to hours (Nakamoto, 2008), while Ethereum's base layer processes roughly 15 to 30 TPS.

By contrast, because Hedera's DAG structure allows nodes to ingest and propagate events in parallel without waiting for global blocks, the primary bottleneck shifts from consensus protocol overhead to raw network bandwidth and local CPU processing limits (Baird, 2016).

Enterprise deployments of Hedera's consensus architecture reliably sustain over 10,000 TPS for token transfers and data integrity operations. Because virtual voting eliminates traditional multi-round vote messaging, transaction finality latencies are cut down to sub-second or low-second intervals (3 to 5 seconds globally), providing deterministic finality where traditional models offer only probabilistic assumptions (Baird et al., 2019).

5.2 Comparative Analysis of Security Guarantees

The core security advantage of Hedera is its mathematical alignment with Asynchronous Byzantine Fault Tolerance (aBFT), the strongest security classification for distributed state machines (Lamport et al., 1982; Baird, 2016).

Many popular consensus protocols rely on partial synchrony models, assuming that network messages will always be delivered within a bounded time frame (Δ). If an adversary delays messages beyond this assumed window, via a coordinated BGP hijacking attack or a targeted network firewall, the system's timing assumptions break down, exposing it to potential

double-spend attacks or liveness stalls (Vukolić, 2015).

Because Hedera's aBFT protocol operates without making any assumptions about network timing, it remains resilient against arbitrary communication delays, network splits, and malicious packet drops (Baird, 2016). As long as the total weight of honest validators stays above the $\frac{2}{3}$ threshold, the network will continue to reach correct, consensus state updates safely and securely.

5.3 Architectural Limitations

While the architectural advantages of the hashgraph framework are apparent, it is essential to highlight key trade-offs and structural limitations:

- **Storage Overhead:** Because nodes continuously gossip about gossip, the local Directed Acyclic Graph expands rapidly over time. Maintaining the historical graph structure requires significant storage capacity, necessitating specialized state-pruning algorithms and mirror-node architectures to offload historical records from core consensus validators (Wang et al., 2020).
- **Bandwidth Intractability:** The gossip protocol requires sustained peer-to-peer data exchange. In networks with thousands of fully independent nodes, the aggregate bandwidth consumption can become a limiting factor, meaning the system works best with well-provisioned, high-capacity network links.
- **Decentralization Mechanics:** Reaching true aBFT finality safely under an open, permissionless system requires robust sybil protection. If anyone can create millions of virtual identities for free, they could easily bypass the $\frac{1}{3}$ fault tolerance threshold. Hedera mitigates this by weighting votes based on economic coin stake (Proof-of-Stake), shifting the security requirement from identity constraints to economic scarcity (Baird et al., 2019).

6. Conclusion and Future Work

Hedera Hashgraph addresses the primary limitations of traditional linear blockchains by replacing sequential block production with a graph-based Directed Acyclic Graph (DAG) and an asynchronous Byzantine Fault Tolerant (aBFT) consensus protocol. By combining a gossip-about-gossip protocol with virtual voting, Hedera eliminates the high network and computational overhead of traditional consensus mechanisms, enabling high transaction throughput, low operational latency, and fair transaction ordering without relying on complex sharding or off-chain scaling layers (Baird, 2016).

Future research in this domain should focus on:

- Designing advanced state-pruning algorithms to reduce the long-term storage requirements of rapidly expanding DAG structures.
- Investigating how leaderless aBFT protocols scale in highly fragmented networks with highly variable bandwidth constraints.
- Optimizing cross-chain interoperability protocols to securely connect DAG-based networks with legacy linear blockchain platforms without compromising their finality speed or security guarantees.

References

- Baird, L. (2016). *The swirls hashgraph consensus algorithm: Fair, fast, byzantine fault tolerance*. Swirls Tech Report SWIRLDS-TR-2016-01.
- Baird, L., Harmon, M., & Madsen, P. (2019). *Hedera: A public hashgraph network & governing council* (Whitepaper). Hedera Hashgraph.
- Bano, S., Sonnino, A., Al-Bassam, M., Azouvi, S., McCorry, P., Meiklejohn, S., & Danezis, G. (2019). SoK: Consensus in the age of blockchains. *Proceedings of the 1st ACM Conference on Advances in Financial Technologies (AFT '19)*, 183–198.
<https://doi.org/10.1145/3318041.3355458>
- Buterin, V. (2017). *The meaning of decentralization*. Medium.
- Castro, M., & Liskov, B. (1999). Practical Byzantine fault tolerance. *Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI '99)*, 173–186.
- DeCandia, G., Hastorun, D., Jampani, M., Kakulapati, G., Lakshman, A., Pilchin, A., Sivasubramanian, S., Voshall, P., & Vogels, W. (2007). Dynamo: Amazon's highly available key-value store. *ACM SIGOPS Operating Systems Review*, 41(6), 205-220.
- King, S., & Nadal, S. (2012). *PPCoin: Peer-to-peer crypto-currency with proof-of-stake*.
- Lamport, L., Shostak, R., & Pease, M. (1982). The Byzantine generals problem. *ACM Transactions on Programming Languages and Systems (TOPLAS)*, 4(3), 382-401.
- Nakamoto, S. (2008). *Bitcoin: A peer-to-peer electronic cash system*. Decentralized Business Review.
- Vukolić, M. (2015). The quest for scalable blockchain fabric: Proof-of-work vs. BFT replication. *International Workshop on Open Problems in Network Security (iNetSec 2015)*, 112-125. Springer.

Wang, Q., Li, R., Wang, Q., & Chen, S. (2020). Exploring the scalability of blockchain-based distributed ledgers: A comprehensive survey. *IEEE Access*, 8, 115002-115032.

Zheng, Z., Xie, S., Dai, H., Chen, X., & Wang, H. (2017). An overview of blockchain technology: Architecture, consensus, and future trends. *2017 IEEE International Congress on Big Data (BigData Congress)*, 557-564.